

Problem Statement and Motivation

Problem Statement: The Duke Chapter of the American Society of Civil Engineers (ASCE) requires a scaled-down concrete canoe design to use as a prototype for a full-sized, four-person model that they will build and race in the ASCE Concrete Canoe Competition in the spring.

Motivation: Our mini-concrete canoe serves as a direct blueprint that Duke ASCE will scale up for their competition canoe.

- Proven design: Validated concrete mix design, structural methods, and construction techniques
- Risk and cost-reduction: Identify and resolve issues at smaller scale before full-scale investment
- Competitive excellence: Enable Duke's return to ASCE competition and demonstrate our leadership in civil engineering innovation

Objectives

- The concrete canoe must float and move quickly and effectively through water during the competition.
- The canoe should appear symmetrical and have a smooth body; it should be aesthetically pleasing to viewers.
- The canoe must be safe for the user and those carrying it – no sharp or rough edges.
- The canoe must be durable, meaning it does not dissolve in water and no major cracks appear in the body after a year of usage.
- The canoe must be comfortable for users who sit in it.

Design Criteria and Testing

Table 1: Summary of Design Objectives and Constraints

	Design Criteria	Target Value/Performance Criteria
Constraints	Buoyancy	Float time after being filled with water ≥ 2 minutes
	Dimensions	Max size: 5.5' x 2' x 3.5'
	Portable	Canoe must be transported to the competition ≥ 4 hours without breaking.
	Made of concrete	Canoe must be composed of concrete
	Safety	No sharp edges or surfaces
Objectives	Durability	Canoe does not dissolve in water Cracks and chips in canoe $< \frac{1}{4}$ in. after a year of use
	Hydrodynamics	Canoe's length to width ratio is within a commonly accepted standard range for stability and speed (slalom and sprints); 4:1 to 6:1
	Aesthetics	Detailing and logo placement ≥ 6 in. above water
	Weight	Weight of concrete ≥ 80.0 PCF
	Ergonomics	Rower should be able to sit ~ 1 hour in the canoe comfortably



Figure 1.1: Concrete mix strength test cubes

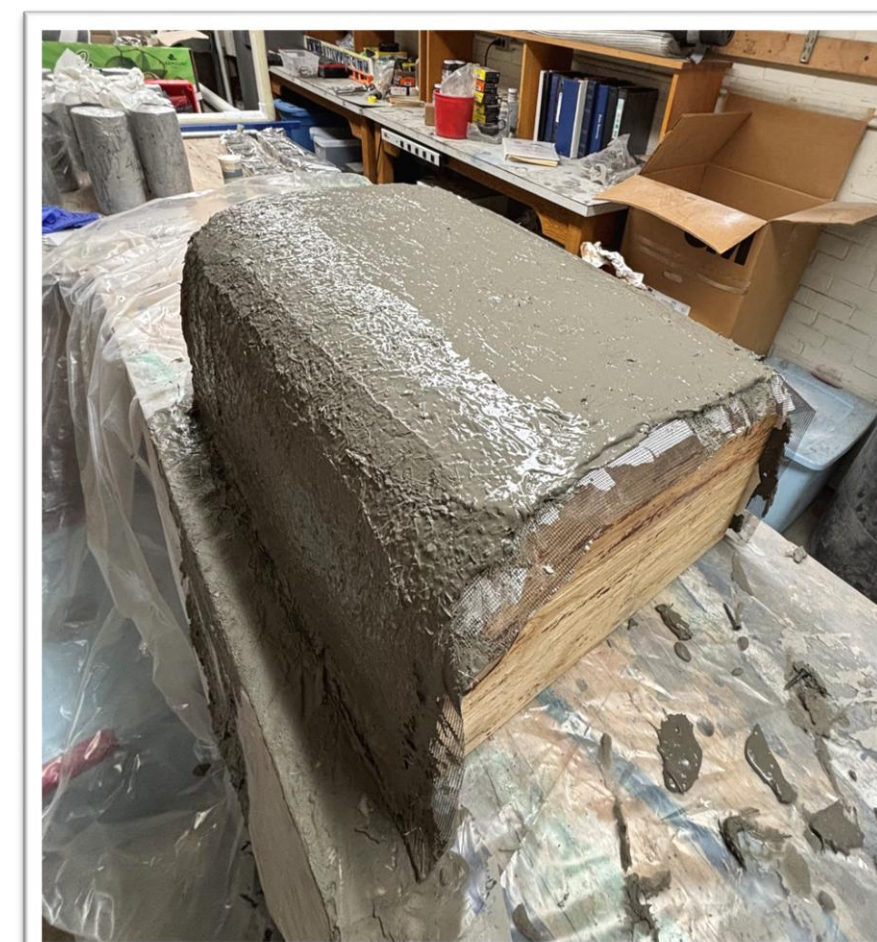


Figure 1.2: Mix workability test

Pictured above on the left are cubes from the concrete mix strength test. We used a hydraulic machine to crush the demo cubes and found that after 28 days it averaged a compressive strength of 909.71 PSI, characteristic of a strong lightweight concrete. Pictured above on the right is a mix workability test which gave us insight into the general constructability of the canoe and the current workability of our mix. Additional tests include a buoyancy test, ergonomics test, and slump test.

Results

How Design Solution Solves Problems:

Our final solution solves the problem by creating a lightweight, durable, and hydrodynamically efficient mini canoe with design features that can be scaled up for the full-sized model. The following design aspects each contribute to shaping our final solution and solving our problem.

- **Round hall bottom:** Allows for a hydrodynamic canoe design while also being extremely durable
- **Chicken wire reinforcement:** Increases the canoe's durability by helping manage tensile forces.
- **No specific seating:** Reduces the canoe's weight by eliminating seating while remaining very ergonomic.
- **Mix design:** Focuses on a lightweight, durable, and water-resistant mix that ensures the canoe floats and races.

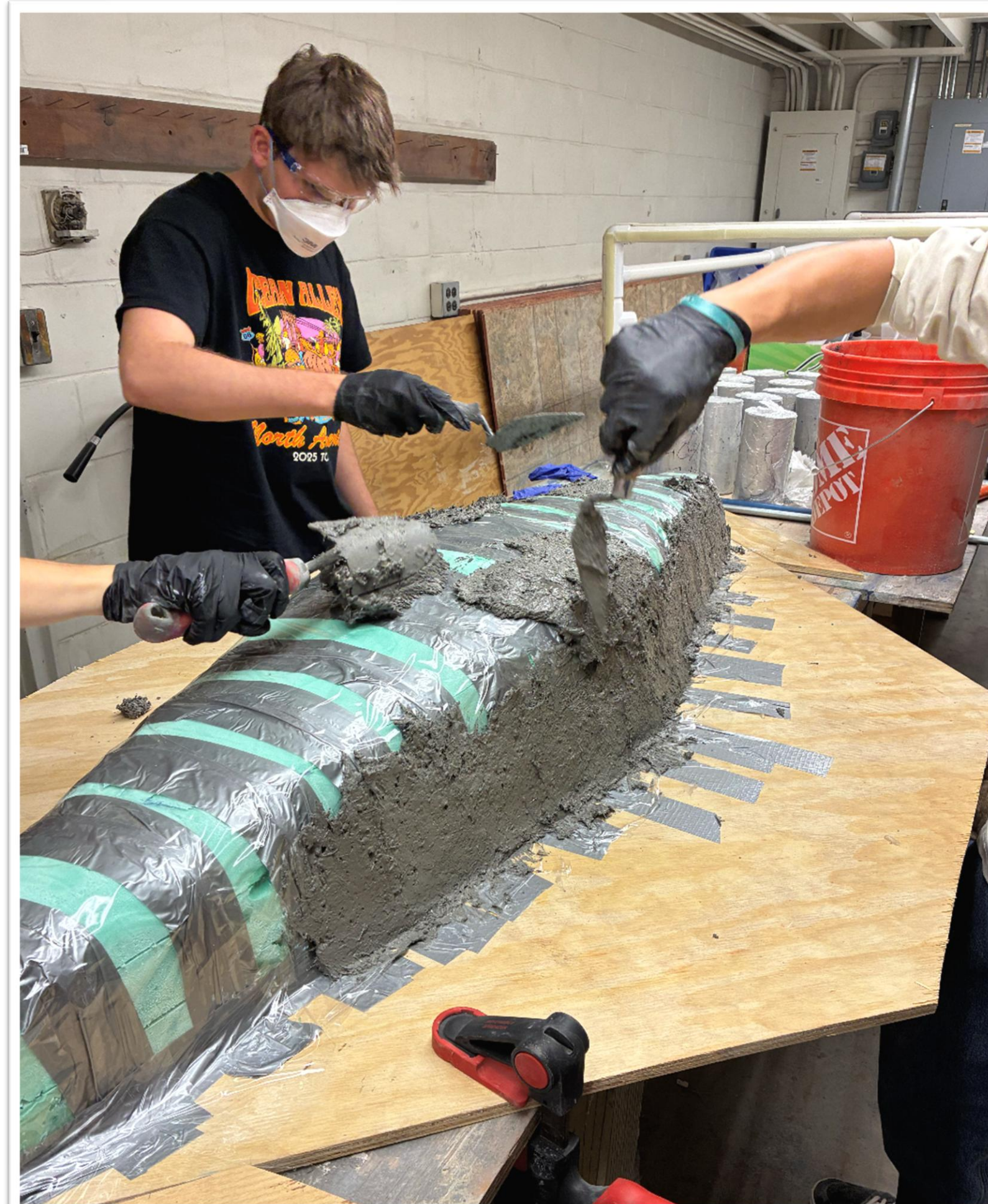


Figure 2.1 : Applying concrete mix to mold



Figure 2.2: Final Canoe



Figure 2.3: Early-stage mix in the concrete mixer

Conclusion and Future Work

Conclusion: Our team successfully developed a comprehensive mini concrete canoe prototype that serves as a direct blueprint for Duke ASCE's competition entry. Through optimized concrete mix design and validated structural methods, we've provided a proven foundation that minimizes risk and maximizes the team's potential for success at competition.

Future Work:

- Scale up design to full-sized 4-person canoe
- Conduct full-scale water and performance testing
- Continue optimization of concrete mix for weight and durability balance

Acknowledgements

Team WWW would like to thank the following individuals for making this possible!

Dr. Laura Dalton, Ph.D., her students Kainā Vieira, Soniya Tiwari, Andrea Bisciotti; Dr. Richard Sheridan, Ph.D.; Professor Stacy Tantom; ASCE clients Oliver O'Donohue, Cory Sands; TAs James Bradley, Hannah Stark; Ali Stocks (Senior Lab Administrator)